# ASSIGNMENT 4

## Q3: Library Management System

**Instructions:** Save JavaScript-only solutions as `.js` files and run with Node.js. For form questions, save the full code as `.html` and open it in a browser. No frameworks are used.

## Solution

```
class Book {
  constructor(title, author, isbn, isIssued = false) {
    Object.assign(this, { title, author, isbn, isIssued });
  }
  issueBook() {
    if (this.isIssued) return `${this.title} is already issued.`;
    this.isIssued = true;
    return `${this.title} has been issued.`;
  }
  returnBook() {
    if (!this.isIssued) return `${this.title} was not issued.`;
    this.isIssued = false;
    return `${this.title} has been returned.`;
  }
}
const books = [
  new Book("The Alchemist", "Paulo Coelho", "ISBN101"),
  new Book("Wings of Fire", "A. P. J. Abdul Kalam", "ISBN102"),
  new Book("1984", "George Orwell", "ISBN103", true)
];
function showAvailableBooks() {
  console.log("Available books:");
  books.filter(book => !book.isIssued)
    .forEach(book => console.log(`${book.title} by ${book.author} (${book.isbn})`));
}
function issueByISBN(isbn) {
  const book = books.find(item => item.isbn === isbn);
  console.log(book ? book.issueBook() : "No book found with that ISBN.");
}
showAvailableBooks();
issueByISBN("ISBN102");
showAvailableBooks();
```

## Expected result / testing

Available books are listed. Issuing ISBN102 marks that book as issued, so it no longer appears in the available list.